

***Statement by Dr Peter Martinez to the UN Secretary-General's Advisory Board on
Disarmament Matters
New York, 24 June 2026***

Transparency and Confidence Building Measures for Outer Space Activities

TCBMs in the space context

Space systems are now part of the critical infrastructure of many states. The rising demand for space-based information and services, coupled with the lowering of the technological and cost entry barriers for space activities over the past twenty years, has led to a sharp increase in the number and diversity of space actors. Consequently, the Earth's orbital environment is becoming congested with active satellites, contaminated by space debris, and contested by rival state and non-state actors alike.

At the same time, space systems have become a critical component of the national military and security capabilities of many states, even those states that are not themselves major space actors but rely on access to the space systems developed by others. Space systems also constitute a critical component of the command-and-control structure of nuclear-armed states and are therefore vital for strategic stability. The growing reliance on space for defence and security creates a new set of vulnerabilities as more states jostle for advantage in the new high ground of space. An increasing number of states are developing, or have already developed, a range of counterspace capabilities that could be used to disrupt, degrade, deceive, or destroy their adversaries' access to space capabilities.

The proliferation of potential counterspace capabilities and corresponding threat vectors have led to much speculation and mistrust among geopolitical rival states about each other's capabilities and intentions in space, giving rise to a situation that is rife with possibilities for misperceptions and miscalculations. Threat perceptions can be very subjective, especially in a climate of mistrust where little information is openly exchanged. States take doctrinal, operational, and institutional measures to mitigate perceived threats, leading to the classic

security dilemma, in which actions taken by one state to increase its own security are perceived by other states as aggressive or threatening, thus producing a vicious spiral that can undermine political and strategic stability in the space domain.

The inherent dual-use character of space technologies makes the security dilemma more acute in space as it is difficult to distinguish between purely civilian space capabilities and the clandestine development or deployment of offensive military space capabilities. Hence, the development of technical capabilities with potential counterspace applications by one state may provoke a sense of insecurity in other states. Because of the rapid pace of technological developments and the lack of agreement on what constitutes a space weapon, achieving meaningful progress on space arms control instruments that focus on regulating technologies or capabilities has been very challenging. In addition to this, there is also no clear, internationally accepted codification of what constitutes responsible behaviour in space.

TCBMs were an inherent part of the policies of cooperative security during the Cold War era and helped to create conditions for later legally binding collective security instruments. As space activities become more prominent in the security and defence policies of states, TCBMs in the space domain are assuming a greater role in fostering regional and global stability.

Characteristics of space TCBMs

There are two categories of space TCBMs – those dealing with capabilities and those dealing with behaviours. Both types of TCBMs help to increase transparency, familiarity and clarity of intentions, and provide a basis for strengthening mutual trust and building confidence amongst states. Both types of TCBMs share the following characteristics.

First, a good TCBM should be clear, practical, and proven, meaning that both the application and the efficacy of the proposed TCBM have been demonstrated by one or more actors.

Second, the application of the proposed TCBM should be objectively verifiable by other parties, either independently or collectively.

Third, the proposed TCBM should reduce or even eliminate the causes of mistrust, misunderstanding, and miscalculation surrounding the space activities of a given state or other space actor.

Past and current multilateral processes for developing space TCBMs

In 2013, a group of governmental experts was mandated by the Secretary-General to study and make recommendations on TCBMs for space activities. This group of experts from fifteen states produced a consensus report that was adopted by the General Assembly without a vote. This report, contained in document A/68/189, proposed a number of pragmatic TCBMs, mostly aimed at increasing transparency through the exchange of information and notifications.

In 2022, pursuant to UNGA General Assembly resolution 76/231, the Open-Ended Working Group on Reducing Space Threats through Norms, Rules and Principles of Responsible Behaviours was convened with a mandate to propose a set of norms, rules, and principles of responsible behaviour in space. Although this Working Group was unable to adopt a consensus report, many of the voluntary and non-binding norms, rules and principles proposed by states as part of this work might similarly be classed as space TCBMs.

The 2024 GGE on further practical measures for the prevention of an arms race in outer space was able to achieve consensus on a report and they had a section on “Elements on transparency and confidence-building measures”. The report also gives a list of possible TCBMs based on the report of 2013 GGE but also the 2023 report of the United Nations Disarmament Commission.

Potential future space TCBMs

So, what could be examples of pragmatic space TCBMs?

One potentially useful TCBM could be about demonstrating commitment to the existing legal framework governing space activities, notably by ratifying the international space treaties; adopting national regulatory frameworks for space activities in conformity with the existing

international legal regime; and by registering space objects, including military objects, in a timely manner in a national registry and with the UN.

A second possible TCBM could be about providing transparency regarding plans and intentions for space activities, including military ones, such as prior notifications of launches, manoeuvres, and close approaches.

A third possible TCBM could be committing to refrain from non-cooperative and uncoordinated rendezvous and close-proximity operations.

A fourth could be sharing information about national military policies, budgets, and programmes pertaining to space.

A fifth could be following existing best practices for orbital debris mitigation, including for military activities.

A sixth could be declaring and acting on a commitment to minimise, as far as practicable, the creation of long-lived orbital debris in the course of normal space operations.

A seventh could be declaring and acting on a commitment not to carry out activities that intentionally generate large amounts of orbital debris.

Although these TCBMs would have greater normative weight if they were to be adopted and implemented multilaterally, there is nothing to prevent a state or group of like-minded states from adopting some or all of these TCBMs on a unilateral basis to demonstrate their commitment to being transparent about their space activities. As more states adopt a given TCBM and implement it consistently over time, indicating that they consider themselves bound by the commitment embodied in that TCBM, the more that TCBM will become an emerging international norm.

Conclusion

TCBMs are an important tool for providing clarity on the application of fundamental principles in the Outer Space Treaty in the context of modern space activities. TCBMs reduce the risk of misperceptions and erroneous assessments of the activities of states in space, thereby fostering regional and global stability. TCBMs in space can take a variety of forms. They may be the elaboration of basic principles related to the exploration and peaceful uses of space, political measures related to establishing norms of conduct, information sharing to improve the transparency of space activities, operational measures which demonstrate a commitment to mutual cooperation in space, and consultative mechanisms aimed at information sharing. Given their non-binding nature, TCBMs complement, but do not substitute for, the verification measures of legally binding agreements and regimes. TCBMs can, however, help to build trust and confidence to negotiate legally binding instruments.

Finally, although non-binding, when widely implemented by the international community, TCBMs can constitute emerging international norms that could themselves be the precursors for future legally binding instruments.

Thank you for your kind attention.